Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Dr. Jorge Duarte (previously Sousa Dairy)

Physical address of dairy:

18005 Williams RD	Hilmar	Merced	95324
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1930

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X022-X032-XXXX X045-X022-X036-XXXX

B. OPERATORS

Duarte, Jorge

Operator name: Duarte, Jorge Telephone no.: Landline (209) 669-6889 Cellular

2545 Emerald Way	Turlock	CA	95382
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Duarte, Jorge

Legal owner name: Duarte, Jorge Telephone no.: Landline (209) 669-6889 Cellular

2545 Emerald Way	Turlock	CA	95382
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	300	0	0	0
Average number	0	0	225	0	0	0
Avg live weight (lbs)	0	0	1,000	0		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 2,352.05 *tons per reporting period*

Total nitrogen from manure: 21,352.50 *lbs per reporting period*

After ammonia losses (30% loss applied): 14,946.75 *lbs per reporting period*

Total phosphorus from manure: 3,613.50 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 4,035,000 *gallons*

Total nitrogen generated: 9,155.44 *lbs*

Total phosphorus generated: 2,949.67 *lbs*

Total potassium generated: 34,008.80 *lbs*

Total salt generated: 138,055.51 *lbs*

	4,035,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	4,035,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - Well #1	Ground water
Irrigation Water - Canal	Surface water
Irrigation Water - Dairy Irrigation Well	Ground water
Irrigation Water - River Water	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	57	57	2	both	X045-X022-X033-XXXX
2	17	17	2	manure	X047-X013-X002-XXXX
3	26	26	1	manure	X045-X022-X069-XXXX
4	18	18	1	none	X024-X009-X200-XXXX X041-X080-X002-XXXX
5	21	21	1	manure	X046-X030-X020-XXXX
Totals for areas that were used for application	121	121	6		
Totals for areas that were not used for application	18	18	1		
Land application area totals	139	139	7		

B. CROPS AND HARVESTS

1

Field name: 1

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 57 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/20/2023	796.00 *ton*	Dry-weight		60.0	19,600.00	4,100.00	30,200.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.96	218.97	45.80	337.39	994.30

1

07/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 57 Plant date: 07/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	1,695.00 *ton*	Dry-weight		66.3	13,800.00	2,200.00	15,000.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.74	276.59	44.09	300.64	1,122.39

2

Field name: 2

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/20/2023	238.00 *ton*	Dry-weight		59.6	19,700.00	4,200.00	29,900.00		8.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.00	222.85	47.51	338.23	950.21

07/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 07/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	497.00 *ton*	Dry-weight		60.6	13,800.00	2,600.00	15,500.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.24	317.92	59.90	357.08	1,359.21

3

Field name: 3

01/01/2011: Almond, in shell

Crop: Almond, in shell Acres planted: 26 Plant date: 01/01/2011

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	32.10 *ton*	Dry-weight		9.3	12,500.00	1,000.00	21,000.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	3.00	390.00	60.00	300.00	0.00
Total actual harvest content	1.23	27.99	2.24	47.03	129.90

4

Field name: 4

01/01/2011: Almond, in shell

Crop: Almond, in shell Acres planted: 18 Plant date: 01/01/2011

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	16.52 *ton*	Dry-weight		10.3	23,100.00	2,300.00	17,300.00		4.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	3.00	390.00	60.00	300.00	0.00
Total actual harvest content	0.92	38.03	3.79	28.48	80.68

5

Field name: 5

5

01/01/2015: Almond, in shell

Crop: Almond, in shell Acres planted: 21 Plant date: 01/01/2015

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	21.96 *ton*	Dry-weight		7.3	18,400.00	1,700.00	19,500.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	3.00	390.00	60.00	300.00	0.00
Total actual harvest content	1.05	35.67	3.30	37.81	110.51

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/20/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	37.22	11.99	138.26	561.24	935,000.00 *gal*
Irrigation Water - River Water	Surface water	0.00	0.00	0.00	38.19	7,050,000.00 *gal*
Application event totals		37.22	11.99	138.26	599.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	199.68	81.45	425.63	0.00	800.00 *ton*
Application event totals		199.68	81.45	425.63	0.00	

1 - 07/15/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 07/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	237.08	109.22	399.58	0.00	800.00 *ton*
Application event totals		237.08	109.22	399.58	0.00	

1 - 07/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	41.00	13.21	152.30	618.26	1,030,000.00 *gal*
Irrigation Water - River Water	Surface water	0.00	0.00	0.00	59.15	10,920,000.00 *gal*
Application event totals		41.00	13.21	152.30	677.41	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.00	0.00	0.00	56.55	10,440,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.55	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	40.60	13.08	150.82	612.26	1,020,000.00 *gal*
Irrigation Water - River Water	Surface water	0.00	0.00	0.00	56.55	10,440,000.00 *gal*
Application event totals		40.60	13.08	150.82	668.81	

09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	41.80	13.47	155.26	630.27	1,050,000.00 *gal*
Irrigation Water - River Water	Surface water	0.00	0.00	0.00	61.43	11,340,000.00 *gal*
Application event totals		41.80	13.47	155.26	691.70	

09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.00	0.00	0.00	61.43	11,340,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.43	

1 - 07/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.00	0.00	0.00	61.43	11,340,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.43	

2 - 11/20/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.00	0.00	0.00	45.77	2,520,000.00 *gal*
Application event totals		0.00	0.00	0.00	45.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	234.33	95.58	499.49	0.00	280.00 *ton*
Application event totals		234.33	95.58	499.49	0.00	

2 - 07/15/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 07/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

2 - 07/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	347.78	160.21	586.15	0.00	350.00 *ton*
Application event totals		347.78	160.21	586.15	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.00	0.00	0.00	56.67	3,120,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.00	0.00	0.00	58.85	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.00	0.00	0.00	58.85	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.00	0.00	0.00	61.03	3,360,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.00	0.00	0.00	61.03	3,360,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.03	

2 - 07/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.00	0.00	0.00	61.03	3,360,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.03	

3 - 01/01/2011: Almond, in shell

Field name: 3

Crop: Almond, in shell

Plant date: 01/01/2011

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	23.89	2,188,875.00 *gal*
Application event totals		0.00	0.00	0.00	23.89	

05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	24.62	2,256,225.00 *gal*
Application event totals		0.00	0.00	0.00	24.62	

05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	26.09	2,390,925.00 *gal*
Application event totals		0.00	0.00	0.00	26.09	

06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	24.99	2,289,900.00 *gal*
Application event totals		0.00	0.00	0.00	24.99	

3 - 01/01/2011: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	24.99	2,289,900.00 *gal*
Application event totals		0.00	0.00	0.00	24.99	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	26.09	2,390,925.00 *gal*
Application event totals		0.00	0.00	0.00	26.09	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	25.72	2,357,250.00 *gal*
Application event totals		0.00	0.00	0.00	25.72	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	24.62	2,256,225.00 *gal*
Application event totals		0.00	0.00	0.00	24.62	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	24.62	2,256,225.00 *gal*
Application event totals		0.00	0.00	0.00	24.62	

10/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	21.89	8.93	46.66	0.00	40.00 *ton*
Application event totals		21.89	8.93	46.66	0.00	

4 - 01/01/2011: Almond, in shell

4 - 01/01/2011: Almond, in shell

Field name: 4

Crop: Almond, in shell

Plant date: 01/01/2011

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Dairy Irrigation Well	Ground water	29.03	0.00	0.00	708.58	1,652,320.00 *gal*
Application event totals		29.03	0.00	0.00	708.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Dairy Irrigation Well	Ground water	17.51	0.00	0.00	427.46	996,780.00 *gal*
Application event totals		17.51	0.00	0.00	427.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Dairy Irrigation Well	Ground water	29.82	0.00	0.00	727.84	1,697,220.00 *gal*
Application event totals		29.82	0.00	0.00	727.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Dairy Irrigation Well	Ground water	17.99	0.00	0.00	439.01	1,023,720.00 *gal*
Application event totals		17.99	0.00	0.00	439.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Dairy Irrigation Well	Ground water	27.45	0.00	0.00	670.07	1,562,520.00 *gal*
Application event totals		27.45	0.00	0.00	670.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Dairy Irrigation Well	Ground water	18.46	0.00	0.00	450.57	1,050,660.00 *gal*
Application event totals		18.46	0.00	0.00	450.57	

5 - 01/01/2015: Almond, in shell

Field name: 5

Crop: Almond, in shell

Plant date: 01/01/2015

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	18.20	1,347,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	17.74	1,313,325.00 *gal*
Application event totals		0.00	0.00	0.00	17.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	17.74	1,313,325.00 *gal*
Application event totals		0.00	0.00	0.00	17.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	16.38	1,212,300.00 *gal*
Application event totals		0.00	0.00	0.00	16.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	17.29	1,279,650.00 *gal*
Application event totals		0.00	0.00	0.00	17.29	

5 - 01/01/2015: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	16.47	1,219,035.00 *gal*
Application event totals		0.00	0.00	0.00	16.47	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	18.20	1,347,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.20	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	19.11	1,414,350.00 *gal*
Application event totals		0.00	0.00	0.00	19.11	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	17.29	1,279,650.00 *gal*
Application event totals		0.00	0.00	0.00	17.29	

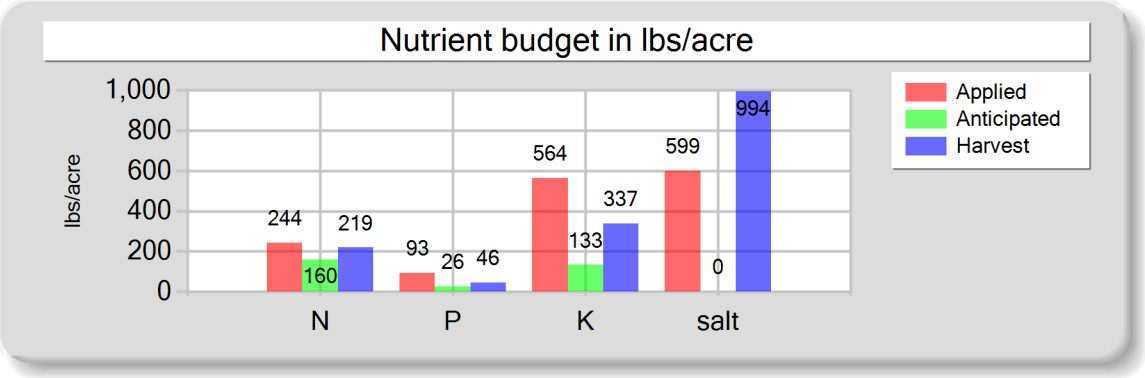
10/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	33.87	13.82	72.21	0.00	50.00 *ton*
Application event totals		33.87	13.82	72.21	0.00	

B. NUTRIENT BUDGET

1 - 11/20/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	199.68	81.45	425.63	0.00
Process wastewater	37.22	11.99	138.26	561.24
Fresh water	0.00	0.00	0.00	38.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	243.90	93.44	563.89	599.43
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	218.97	45.80	337.39	994.30
Nutrient balance	24.93	47.63	226.50	-394.87
Applied to removed ratio	1.11	2.04	1.67	0.60

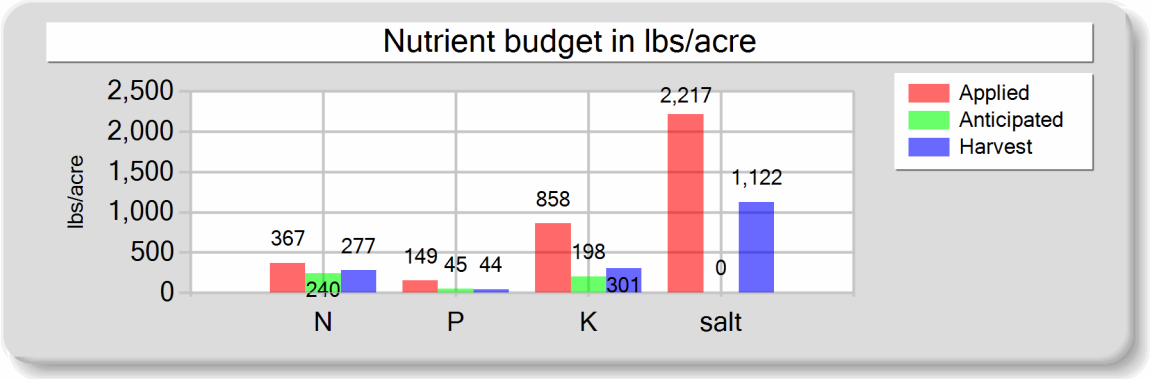
Fresh water applied	
7,050,000.00	*gallons*
259.63	*acre-inches*
4.55	*inches/acre*

Process wastewater applied	
935,000.00	*gallons*
34.43	*acre-inches*
0.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 07/15/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 07/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	237.08	109.22	399.58	0.00
Process wastewater	123.40	39.76	458.39	1,860.79
Fresh water	0.00	0.00	0.00	356.54
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	367.49	148.98	857.97	2,217.33
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	276.59	44.09	300.64	1,122.39
Nutrient balance	90.90	104.88	557.33	1,094.94
Applied to removed ratio	1.33	3.38	2.85	1.98

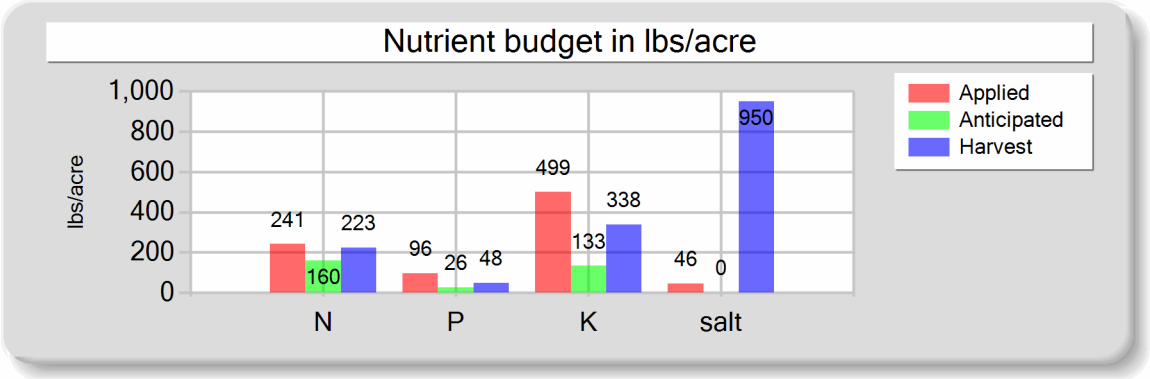
Fresh water applied	
65,820,000.00	*gallons*
2,423.93	*acre-inches*
42.53	*inches/acre*

Process wastewater applied	
3,100,000.00	*gallons*
114.16	*acre-inches*
2.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/20/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	234.33	95.58	499.49	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	45.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	241.33	95.58	499.49	45.77
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	222.85	47.51	338.23	950.21
Nutrient balance	18.48	48.07	161.26	-904.44
Applied to removed ratio	1.08	2.01	1.48	0.05

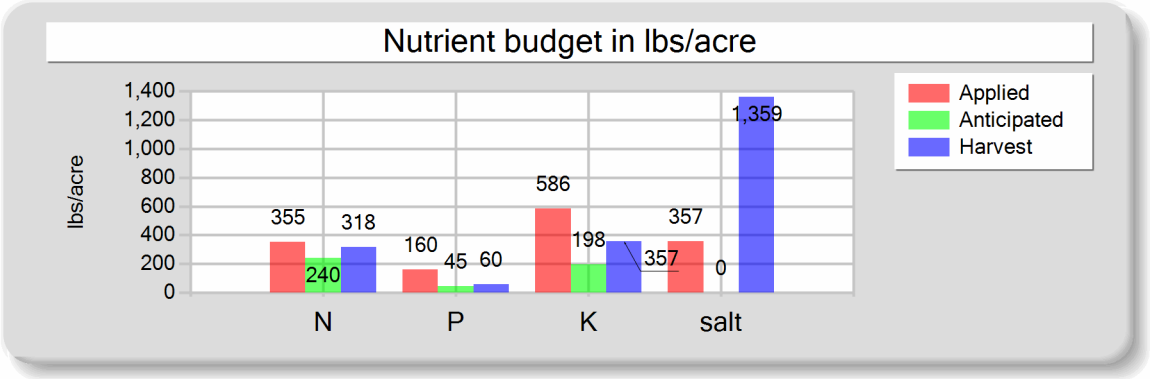
Fresh water applied	
2,520,000.00	*gallons*
92.80	*acre-inches*
5.46	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 07/15/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 07/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	347.78	160.21	586.15	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	357.44
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	354.78	160.21	586.15	357.44
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	317.92	59.90	357.08	1,359.21
Nutrient balance	36.86	100.32	229.07	-1,001.77
Applied to removed ratio	1.12	2.67	1.64	0.26

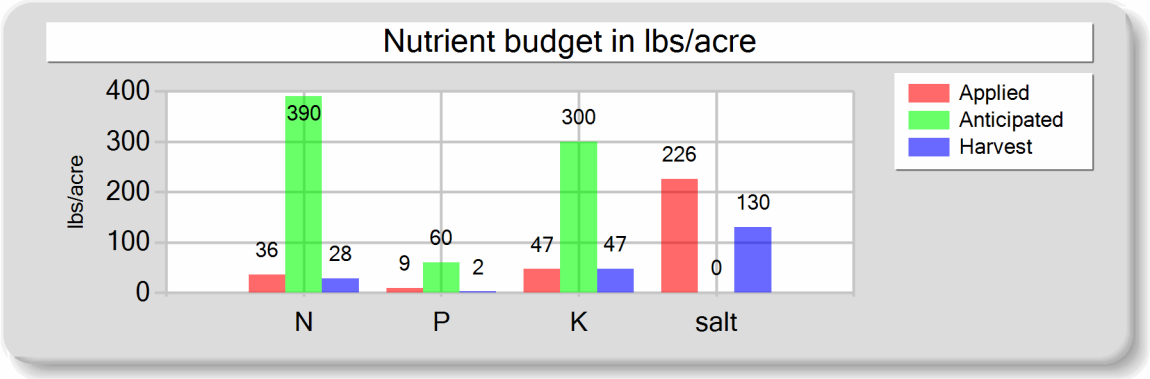
Fresh water applied	
19,680,000.00	*gallons*
724.75	*acre-inches*
42.63	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 01/01/2011: Almond, in shell

Field name: 3 Crop: Almond, in shell Plant date: 01/01/2011



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	21.89	8.93	46.66	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	225.64
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	35.89	8.93	46.66	225.64
Anticipated crop nutrient removal	390.00	60.00	300.00	0.00
Actual crop nutrient removal	27.99	2.24	47.03	129.90
Nutrient balance	7.89	6.69	-0.38	95.74
Applied to removed ratio	1.28	3.99	0.99	1.74

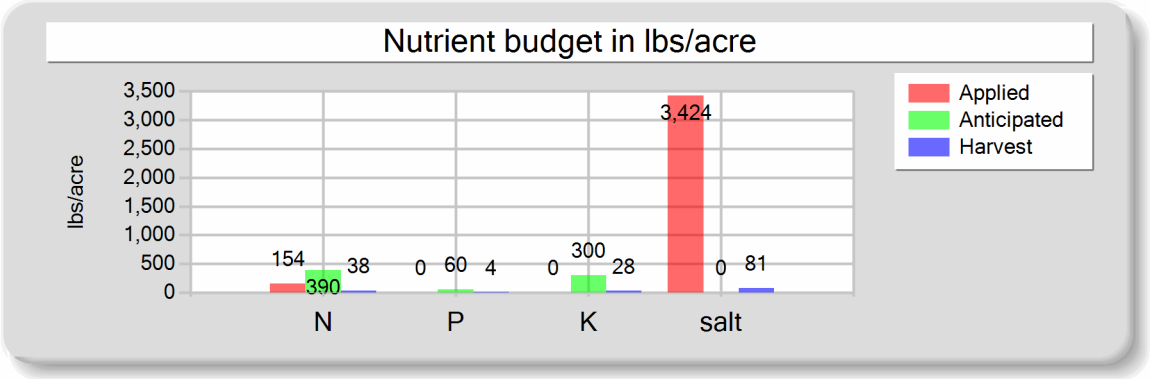
Fresh water applied	
20,676,450.00	*gallons*
761.44	*acre-inches*
29.29	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 01/01/2011: Almond, in shell

Field name: 4 Crop: Almond, in shell Plant date: 01/01/2011



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	140.27	0.00	0.00	3,423.53
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	154.27	0.00	0.00	3,423.53
Anticipated crop nutrient removal	390.00	60.00	300.00	0.00
Actual crop nutrient removal	38.03	3.79	28.48	80.68
Nutrient balance	116.24	-3.79	-28.48	3,342.85
Applied to removed ratio	4.06	0.00	0.00	42.43

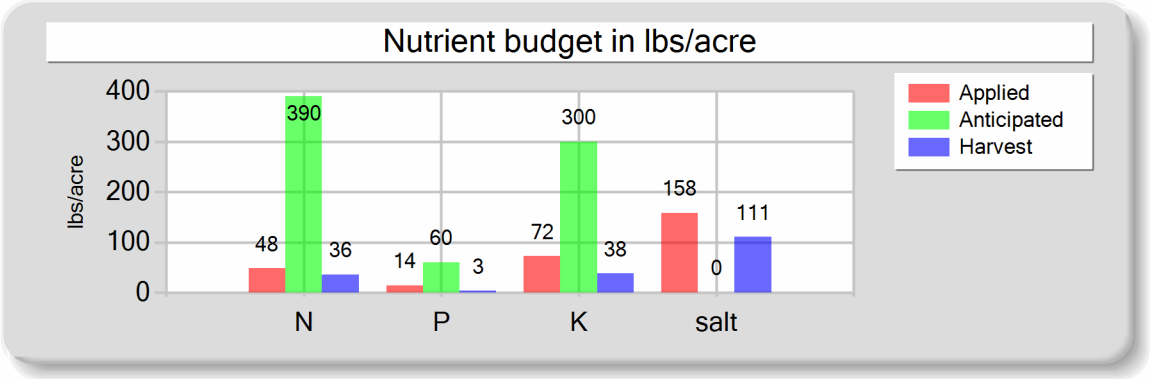
Fresh water applied	
7,983,220.00	*gallons*
293.99	*acre-inches*
16.33	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 01/01/2015: Almond, in shell

Field name: 5 Crop: Almond, in shell Plant date: 01/01/2015



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	33.87	13.82	72.21	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	158.42
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	47.87	13.82	72.21	158.42
Anticipated crop nutrient removal	390.00	60.00	300.00	0.00
Actual crop nutrient removal	35.67	3.30	37.81	110.51
Nutrient balance	12.20	10.52	34.40	47.92
Applied to removed ratio	1.34	4.19	1.91	1.43

Fresh water applied	
11,725,635.00	*gallons*
431.82	*acre-inches*
20.56	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure
Sample date: 05/25/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 5.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,900.00	4,100.00	15,000.00							0.00
DL	100.00	100.00	100.00							1.00

Solid Manure

Sample and source description: Solid Manure
Sample date: 11/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 6.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,600.00	3,100.00	16,200.00							0.00
DL	100.00	100.00	100.00							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water
Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	271.00	110.00	0.00	0.90	87.60	1,010.00								5,410.00	4,100
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	192.00	88.90	0.00	0.50	101.00	940.00								5,220.00	3,190
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	80.90	45.30	0.00	0.50	38.70	328.00								1,950.00	1,550
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	102.00	38.30	0.00	0.70	33.40	331.00								1,870.00	1,550
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - Well #1

Domestic Well - Well #1

Well #1

Sample description: Well #1

Sample date: 11/05/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			0.30								850.00	
DL			0.10								10.00	

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 07/24/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00										31.90	34
DL	1.00										10.00	10

Irrigation Water - Dairy Irrigation Well

Dairy Irrigation Well

Sample description: Dairy Irrigation Well

Sample date: 07/24/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	37.90										1,480.00	925
DL	1.00										10.00	10

Irrigation Water - River Water

Irrigation Water - River Water

River Water

Sample description: River Water

Sample date: 07/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00										35.20	37
DL	1.00										10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 11/20/2022: Oats, silage-soft dough

Dairy Field A

Sample and source description: Dairy Field A

Sample date: 04/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,600.00	4,100.00	30,200.00		8.90
DL	100.00	100.00	100.00		0.01

1 - 07/15/2023: Corn, silage

1 - 07/15/2023: Corn, silage

6A

Sample and source description: 6A

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	2,200.00	15,000.00		5.60
DL	100.00	100.00	100.00		0.01

2 - 11/20/2022: Oats, silage-soft dough

Dairy Field B

Sample and source description: Dairy Field B

Sample date: 04/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,700.00	4,200.00	29,900.00		8.40
DL	100.00	100.00	100.00		0.01

2 - 07/15/2023: Corn, silage

6B

Sample and source description: 6B

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	2,600.00	15,500.00		5.90
DL	100.00	100.00	100.00		0.01

3 - 01/01/2011: Almond, in shell

W-N-S-C #2

Sample and source description: W-N-S-C #2

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 9.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	1,000.00	21,000.00		5.80
DL	100.00	100.00	100.00		0.01

4 - 01/01/2011: Almond, in shell

W-N-S-C #1

Sample and source description: W-N-S-C #1

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,100.00	2,300.00	17,300.00		4.90
DL	100.00	100.00	100.00		0.01

5 - 01/01/2015: Almond, in shell

W-N-S-C #3

Sample and source description: W-N-S-C #3

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	1,700.00	19,500.00		5.70
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

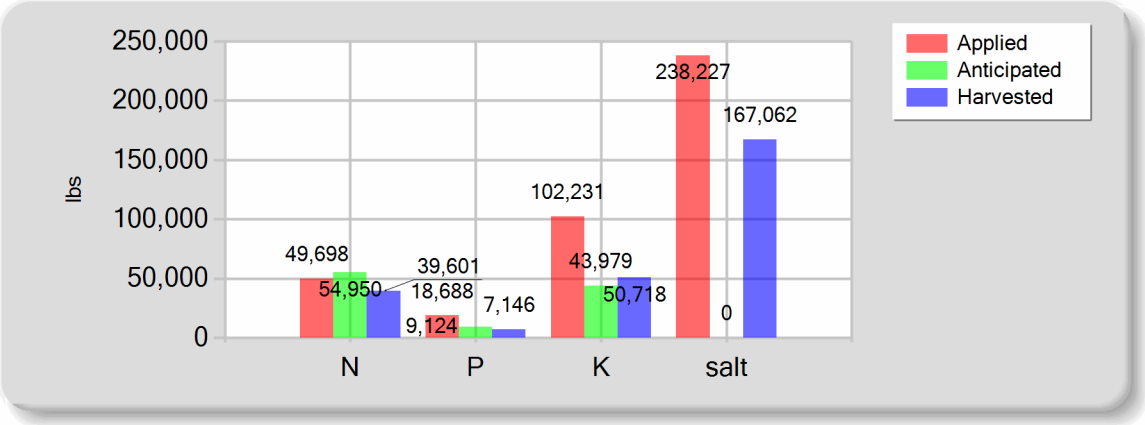
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

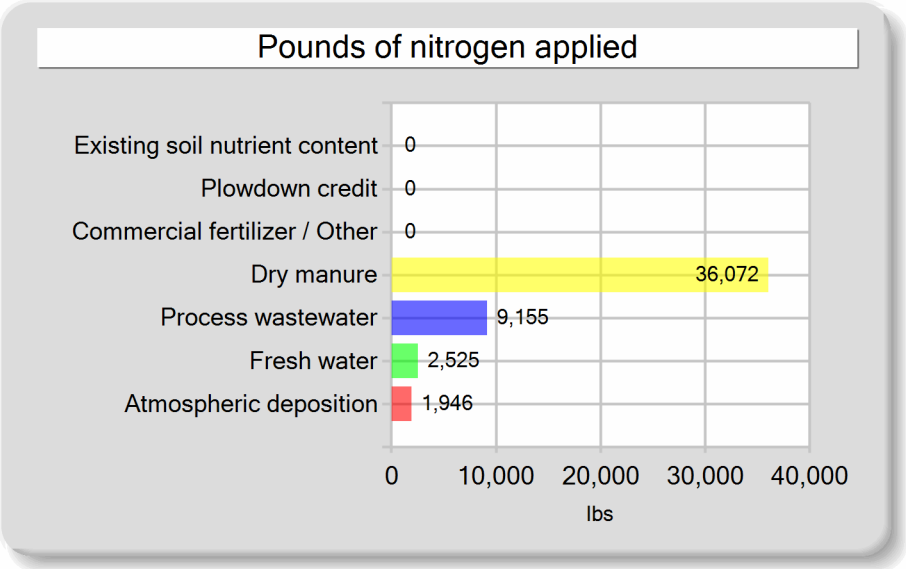
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

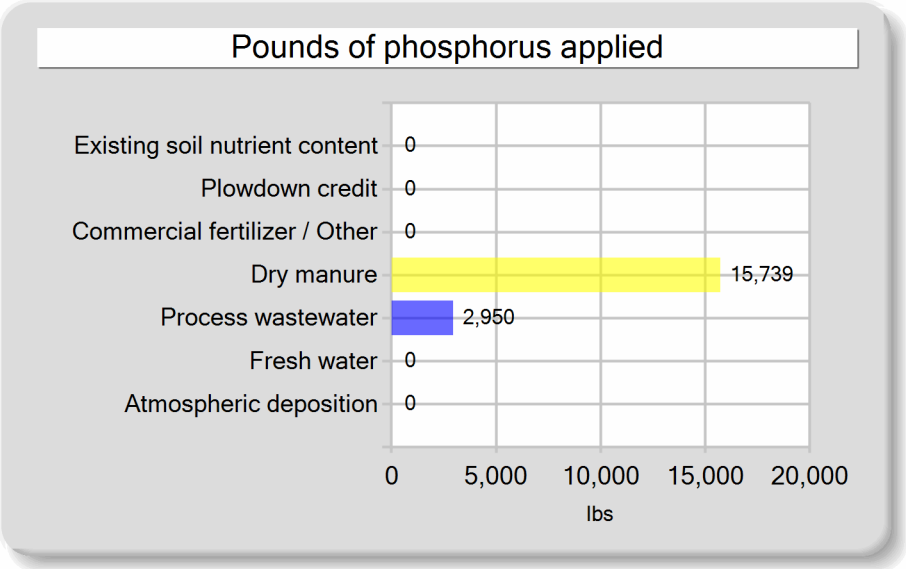
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	36,071.85	15,738.81	68,222.39	0.00
Process wastewater	9,155.44	2,949.67	34,008.80	138,055.51
Fresh water	2,524.90	0.00	0.00	100,171.21
Atmospheric deposition	1,946.00	0.00	0.00	0.00
Total nutrients applied	49,698.19	18,688.49	102,231.18	238,226.71
Anticipated crop nutrient removal	54,950.00	9,124.40	43,979.20	0.00
Actual crop nutrient removal	39,601.39	7,145.76	50,717.51	167,061.54
Nutrient balance	10,096.79	11,542.72	51,513.67	71,165.17
Applied to removed ratio	1.25	2.62	2.02	1.43

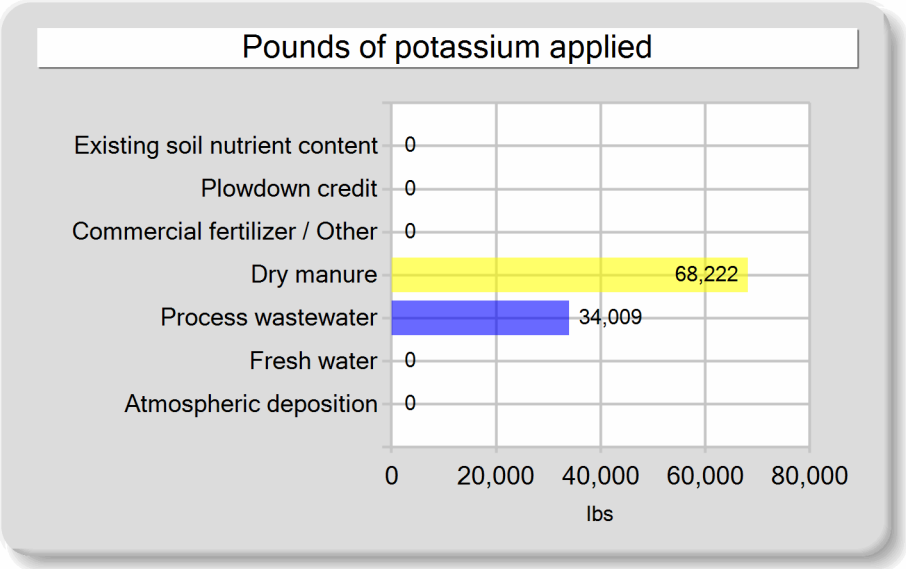
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

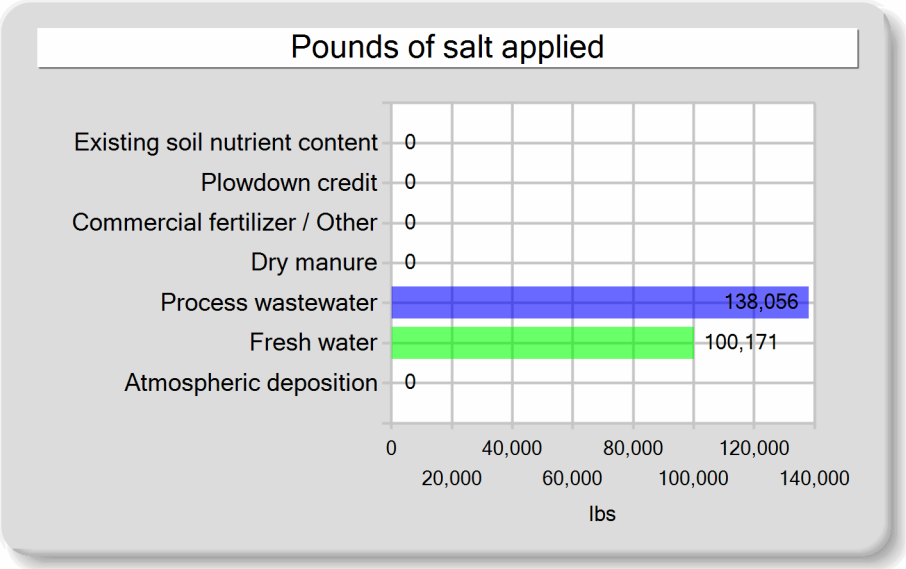


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Dr. Jorge Duarte
Dr. Jorge Duarte (Jun 20, 2024 12:43 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Jorge Duarte	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
06/20/2024	
DATE	DATE

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Joe O. Rocha

Received: 07/25/2023 16:30
Reported: 07/31/2023 10:28

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23G1768-01	Canal	Ag Water	T. Madruga		07/24/2023 15:00
23G1768-02	Dairy Irrigation Well	Ag Water	T. Madruga		07/24/2023 16:00

Default Cooler Temperature on Receipt °C: -3.8
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Sample Results

Sample: Canal
23G1768-01 (Water)

Sampled: 7/24/2023 15:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.03**	mmhos/cm	0.01	1		07/28/23 12:08	SM 2510 B		BEG0955
Electrical Conductivity umhos	**31.9**	umhos/cm	10.0	1		07/28/23 12:08	SM 2510 B		BEG0955
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	07/25/23 20:14	EPA 300.0		BEG0912
pH	**6.7**	units	1.0	1		07/28/23 12:08	SM 4500-H+	H	BEG0955
Total Filterable Solids (TDS)	**34.0**	mg/L	10.0	1		07/28/23 15:53	SM 2540 C		BEG0941
Temperature	**25.0**	°C	0.0	1		07/28/23 12:08	SM 2510 B		BEG0955
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		07/27/23 15:20	SM 4500-NH3 C		BEG0965
Total Nitrogen	ND	mg/L	1.00	1		07/27/23 15:20	SM 4500-NH3 C		BEG0965

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Sample Results
(Continued)

Sample: Dairy Irrigation Well
23G1768-02 (Water)

Sampled: 7/24/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.48**	mmhos/cm	0.01	1		07/28/23 12:09	SM 2510 B		BEG0955
Electrical Conductivity umhos	**1480**	umhos/cm	10.0	1		07/28/23 12:09	SM 2510 B		BEG0955
Nitrate Nitrogen as NO3N	**37.0**	mg/L	0.1	1	10	07/25/23 20:35	EPA 300.0		BEG0912
pH	**7.1**	units	1.0	1		07/28/23 12:09	SM 4500-H+	H	BEG0955
Total Filterable Solids (TDS)	**925**	mg/L	10.0	1		07/28/23 15:53	SM 2540 C		BEG0941
Temperature	**25.0**	°C	0.0	1		07/28/23 12:09	SM 2510 B		BEG0955
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		07/27/23 15:21	SM 4500-NH3 C		BEG0965
Total Nitrogen	**37.9**	mg/L	1.00	1		07/27/23 15:21	SM 4500-NH3 C		BEG0965

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Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG0912										
Blank (BEG0912-BLK1)					Prepared & Analyzed: 7/25/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEG0912-BLK2)					Prepared & Analyzed: 7/25/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEG0912-BLK3)					Prepared: 7/25/2023 Analyzed: 7/26/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEG0912-BS1)					Prepared: 7/25/2023 Analyzed: 7/26/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.1	90-110		
LCS (BEG0912-BS2)					Prepared: 7/25/2023 Analyzed: 7/26/2023					
Nitrate Nitrogen as NO3N	4.8		0.1	mg/L	5.000		96.3	90-110		
Duplicate (BEG0912-DUP1)		**Source: 23G1769-01**			Prepared: 7/25/2023 Analyzed: 7/26/2023					
Nitrate Nitrogen as NO3N	3.7		0.1	mg/L		3.6			1.18	10
Duplicate (BEG0912-DUP2)		**Source: 23G1769-03**			Prepared: 7/25/2023 Analyzed: 7/26/2023					
Nitrate Nitrogen as NO3N	2.0		0.1	mg/L		2.0			1.55	10
Matrix Spike (BEG0912-MS1)		**Source: 23G1769-01**			Prepared & Analyzed: 7/25/2023					
Nitrate Nitrogen as NO3N	8.6		0.1	mg/L	5.000	3.6	98.8	90-110		
Matrix Spike (BEG0912-MS2)		**Source: 23G1769-03**			Prepared: 7/25/2023 Analyzed: 7/26/2023					
Nitrate Nitrogen as NO3N	6.9		0.1	mg/L	5.000	2.0	97.4	90-110		
Reference (BEG0912-SRM1)					Prepared & Analyzed: 7/25/2023					
Nitrate Nitrogen as NO3N	9.6			mg/L	10.00		95.5	90-110		
Reference (BEG0912-SRM2)					Prepared: 7/25/2023 Analyzed: 7/26/2023					
Nitrate Nitrogen as NO3N	9.6			mg/L	10.00		95.7	90-110		
Reference (BEG0912-SRM3)					Prepared: 7/25/2023 Analyzed: 7/26/2023					
Nitrate Nitrogen as NO3N	9.6			mg/L	10.00		95.8	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG0941										
Blank (BEG0941-BLK1)					Prepared: 7/25/2023	Analyzed: 7/28/2023				
Total Filterable Solids (TDS)	ND		10.0	mg/L						
LCS (BEG0941-BS1)					Prepared: 7/25/2023	Analyzed: 7/28/2023				
Total Filterable Solids (TDS)	37.5		10.0	mg/L	2000		1.88	0-200		
Duplicate (BEG0941-DUP1)		**Source: 23G1523-01**			Prepared: 7/25/2023	Analyzed: 7/28/2023				
Total Filterable Solids (TDS)	2830		10.0	mg/L		2750			2.87	10
Duplicate (BEG0941-DUP2)		**Source: 23G1819-02**			Prepared: 7/25/2023	Analyzed: 7/28/2023				
Total Filterable Solids (TDS)	313		10.0	mg/L		337			7.18	10
Reference (BEG0941-SRM1)					Prepared: 7/25/2023	Analyzed: 7/28/2023				
Total Filterable Solids (TDS)	353			mg/L	325.0		109	90-110		
Reference (BEG0941-SRM2)					Prepared: 7/25/2023	Analyzed: 7/28/2023				
Total Filterable Solids (TDS)	530			mg/L	495.0		107	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG0955										
Blank (BEG0955-BLK1)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.2		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEG0955-BLK2)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.4		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEG0955-BLK3)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
pH	5.4		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEG0955-DUP1)		**Source: 23G1769-02**			Prepared: 7/26/2023	Analyzed: 7/28/2023				
pH	7.8		1.0	units		7.9			0.508	10
Electrical Conductivity	0.15		0.01	mmhos/cm		0.16			1.36	10
Electrical Conductivity umhos	153		10.0	umhos/cm		155			1.36	10
Duplicate (BEG0955-DUP2)		**Source: 23G1805-08**			Prepared: 7/26/2023	Analyzed: 7/28/2023				
pH	6.9		1.0	units		6.9			0.289	10
Electrical Conductivity	0.03		0.01	mmhos/cm		0.03			0.331	10
Electrical Conductivity umhos	30.2		10.0	umhos/cm		30.3			0.331	10
Reference (BEG0955-SRM1)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
Electrical Conductivity	547			umhos/cm	538.0		102	90-110		
Reference (BEG0955-SRM2)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
pH	5.9			units	5.820		101	28178-101.71		
Reference (BEG0955-SRM3)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEG0955-SRM4)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEG0955-SRM5)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		

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Ranch: Joe O. Rocha

Received: 07/25/2023 16:30
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG0955 (Continued)										
Reference (BEG0955-SRM5)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEG0955-SRM6)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
pH	4.1			units	4.000		102	97.5-102.5		
Reference (BEG0955-SRM7)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
pH	4.1			units	4.000		102	97.5-102.5		
Reference (BEG0955-SRM8)					Prepared: 7/26/2023	Analyzed: 7/28/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG0965										
Blank (BEG0965-BLK1)					Prepared: 7/26/2023	Analyzed: 7/27/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEG0965-BLK2)					Prepared: 7/26/2023	Analyzed: 7/27/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEG0965-BS1)					Prepared: 7/26/2023	Analyzed: 7/27/2023				
Kjeldahl Nitrogen (TKN), Total	5.63		1.00	mg/L	5.709		98.6	90-110		
LCS (BEG0965-BS2)					Prepared: 7/26/2023	Analyzed: 7/27/2023				
Kjeldahl Nitrogen (TKN), Total	5.85		1.00	mg/L	5.709		103	90-110		
Duplicate (BEG0965-DUP1)		**Source: 23G1765-01**			Prepared: 7/26/2023	Analyzed: 7/27/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Duplicate (BEG0965-DUP2)		**Source: 23G1821-02**			Prepared: 7/26/2023	Analyzed: 7/27/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Matrix Spike (BEG0965-MS1)		**Source: 23G1765-01**			Prepared: 7/26/2023	Analyzed: 7/27/2023				
Kjeldahl Nitrogen (TKN), Total	8.87		1.40	mg/L	7.992	ND	111	90-110		
Matrix Spike (BEG0965-MS2)		**Source: 23G1821-02**			Prepared: 7/26/2023	Analyzed: 7/27/2023				
Kjeldahl Nitrogen (TKN), Total	5.23		1.40	mg/L	7.992	ND	65.5	90-110		
Reference (BEG0965-SRM1)					Prepared: 7/26/2023	Analyzed: 7/27/2023				
Kjeldahl Nitrogen (TKN), Total	24.7			mg/L	23.80		104	90-110		

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07/25/23 16:30 23G1768

RG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 2 No. of Bottles:

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Results Need By

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State: CA** **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT Joe O. Rocha
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

( ) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
(X) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Canal	7/24/2023	1500	~3.8°C	
2	Dairy Irrigation Well	7/24/2023	1600	~3.3°C	
3					
4					
5					
6					
7					
8					
9					
10					

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	7/24/2023 @ 1600	7/25/2023 @ 930
Second		Dellavalle Laboratory, Inc.	7/25/2023 @ 930	7/25 3:40
Third	RG	DLI	25 July 16:30	
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

07/25/23 16:30 23G1768

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☐ Walk In ☐ DLI Sampler ☒
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1								
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Dr. Duarte

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:21

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0901-01	Well #1	Ag Water	T. Madruga		11/15/2023 16:00

Default Cooler Temperature on Receipt °C: 2.2
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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LABORATORY INC

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Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Dr. Duarte

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:21

Sample Results

Sample: Well #1
23K0901-01 (Water)

Sampled: 11/15/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.85**	mmhos/cm	0.01	1		11/20/23 14:25	SM 2510 B		BEK0617
Electrical Conductivity umhos	**850**	umhos/cm	10.0	1		11/20/23 14:25	SM 2510 B		BEK0617
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 16:00	Field		BEK0561
Nitrate Nitrogen as NO3N	**0.3**	mg/L	0.1	1	10	11/17/23 00:35	EPA 300.0		BEK0573
Temperature	**25.0**	units	0.0	1		11/20/23 14:25	SM 4500-H+	H	BEK0617
pH	**8.0**	units	1.0	1		11/20/23 14:25	SM 4500-H+	H	BEK0617

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Dr. Duarte

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:21

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0573										
Blank (BEK0573-BLK1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0573-BLK2)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0573-BLK3)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0573-BLK4)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0573-BS1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
LCS (BEK0573-BS2)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		104	90-110		
LCS (BEK0573-BS3)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
Duplicate (BEK0573-DUP1)		**Source: 23K0896-02**			Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	5.8		0.1	mg/L		5.8			0.173	10
Duplicate (BEK0573-DUP2)		**Source: 23K0901-01**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	0.3		0.1	mg/L		0.3			0.629	10
Duplicate (BEK0573-DUP3)		**Source: 23K0935-02**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	1.5		0.1	mg/L		1.5			0.195	10
Matrix Spike (BEK0573-MS1)		**Source: 23K0896-02**			Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	11.0		0.1	mg/L	5.000	5.8	104	90-110		
Matrix Spike (BEK0573-MS2)		**Source: 23K0901-01**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000	0.3	97.1	90-110		
Matrix Spike (BEK0573-MS3)		**Source: 23K0935-02**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	6.8		0.1	mg/L	5.000	1.5	106	90-110		
Reference (BEK0573-SRM1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEK0573-SRM2)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Dr. Duarte

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:21

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0573 (Continued)										
Reference (BEK0573-SRM3)					Prepared: 11/16/2023	Analyzed: 11/17/2023				
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0573-SRM4)					Prepared: 11/16/2023	Analyzed: 11/17/2023				
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Dr. Duarte

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:21

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0617										
Blank (BEK0617-BLK1)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.7		1.0	units						
Blank (BEK0617-BLK2)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.6		1.0	units						
Blank (BEK0617-BLK3)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.0		1.0	units						
Duplicate (BEK0617-DUP1)		**Source: 23K0902-03**			Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	0.79		0.01	mmhos/cm		0.78			0.800	10
pH	7.7		1.0	units		7.7			0.130	10
Electrical Conductivity umhos	791		10.0	umhos/cm		785			0.800	10
Duplicate (BEK0617-DUP2)		**Source: 23K0904-01**			Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1.59		0.01	mmhos/cm		1.59			0.00	10
Electrical Conductivity umhos	1590		10.0	umhos/cm		1590			0.00	10
pH	8.0		1.0	units		8.0			0.125	10
Reference (BEK0617-SRM1)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	555			umhos/cm	538.0		103	90-110		
Reference (BEK0617-SRM2)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	7.6			units	7.520		101	67021-101.32		
Reference (BEK0617-SRM3)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEK0617-SRM4)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEK0617-SRM5)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Dr. Duarte

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:21

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0617 (Continued)										
Reference (BEK0617-SRM5)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEK0617-SRM6)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0617-SRM7)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0617-SRM8)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		

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11/16/23 15:35 23K0901

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 1 No. of Bottles: 1

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Results Need By

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State: CA** **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:
REQUESTED BY: Rui & John Avila
PROJECT XXXXXXXXXXXXXXXX Dr. Duarte
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:
Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Well #1	11/15/2023	1600	2.2	ND
2					
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/15/2023 @ 1600	11/16/2023 @9am
Second	JLF	Dellavalle Laboratory, Inc.	11/16/2023 @9am	11-16-23 3:25
Third	~~DCL~~	DCL	11/16/23 15:35	
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
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Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/16/23 15:35 23K0901

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☑ None ☐ Box ☐

Refrigerant: Wet Ice:☑ Blue Ice:☐ None:☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1									
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										